

HDOC DAY-38

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Batch-15

Question Statement

Write a menu-driven C program using separate functions to:

1. Count spaces, digits and special characters in a string.
2. Count the frequency of a given character.
3. Toggle the case of each character in a string.
4. Exit.

Algorithm

1. Start.
2. Input a string from the user.
3. Display the menu:
 - Count spaces, digits and special characters.
 - Count frequency of a character.
 - Toggle case of characters.
 - Exit.
4. Read the user's choice.
5. Call the corresponding function.
6. Display the result.
7. Repeat the menu until the user selects **Exit**.
8. Stop.

Test Case	Input String	Choice	Expected Output
1	Hello 123!	1	Spaces = 1, Digits = 3, Special = 1
2	Hello World	2, 1	Frequency = 3
3	Hello World	3	hELLO wORLD
4	—	4	Program exits

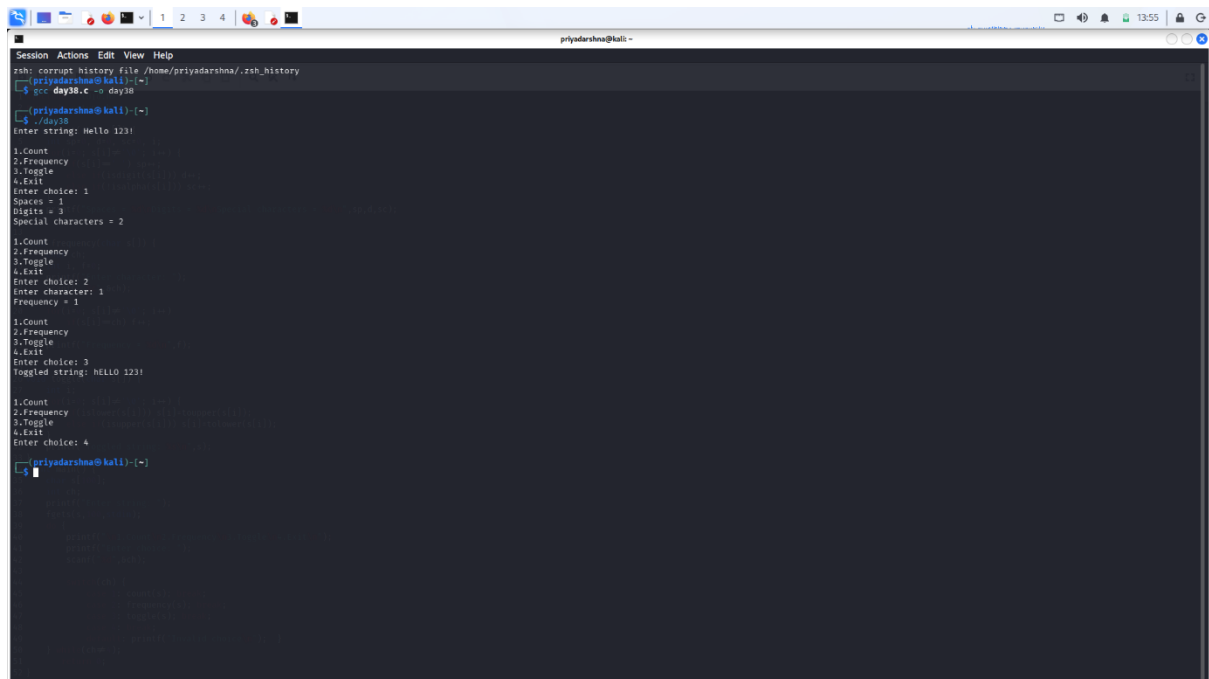
>C PROGRAM:

```

1 #include <stdio.h>
2 #include <ctype.h>
3
4 void count(char s[]) {
5     int sp=0, d=0, sc=0, i;
6     for(i=0; s[i]!='\0'; i++) {
7         if(s[i]==' ') sp++;
8         else if(isdigit(s[i])) d++;
9         else if(isalpha(s[i])) sc++;
10    }
11    printf("Spaces = %d\Digits = %d\Special characters = %d\n",sp,d,sc);
12}
13
14 void frequency(char s[]) {
15     char ch;
16     int i, f=0;
17     printf("Enter character: ");
18     scanf("%c",&ch);
19
20     for(i=0; s[i]!='\0'; i++)
21         if(s[i]==ch) f++;
22
23     printf("Frequency = %d\n",f);
24}
25
26 void toggle(char s[]) {
27     int i;
28     for(i=0; s[i]!='\0'; i++) {
29         if(islower(s[i])) s[i]=toupper(s[i]);
30         else if(isupper(s[i])) s[i]=tolower(s[i]);
31     }
32     printf("Toggled string: %s\n",s);
33}
34
35 int main() {
36     char s[100];
37     int ch;
38     printf("Enter string: ");
39     fgets(s,100,stdin);
40
41     do {
42         printf("\n1.Count\n2.Frequency\n3.Toggle\n4.Exit\n");
43         printf("Enter choice: ");
44         scanf("%d",&ch);
45
46         switch(ch) {
47             case 1: count(s); break;
48             case 2: frequency(s); break;
49             case 3: toggle(s); break;
50             case 4: break;
51             default: printf("Invalid choice\n"); }
52     } while(ch!=4);
53     return 0;
54}

```

>OUTPUT:



```
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
~ [priyadarshna@kali: ~]$ gcc day38.c -o day38
~ [priyadarshna@kali: ~]$ ./day38
Enter string: Hello 123!
1.Count
2.Frequency
3.Toggle
4.Exit
Enter choice: 1
Spaces = 1
Digits = 3
Special characters = 2
1.Count
2.Frequency
3.Toggle
4.Exit
Enter choice: 2
Enter character: 1
Frequency = 1
1.Count
2.Frequency
3.Toggle
4.Exit
Enter choice: 3
Toggled string: HELLO 123!
1.Count
2.Frequency
3.Toggle
4.Exit
Enter choice: 4
~ [priyadarshna@kali: ~]$
```